Lighting

Informática Gráfica



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Grado en Ingeniería Informática

Graphics and Imaging Lab

Geometric optics

Geometric optics

Assumptions

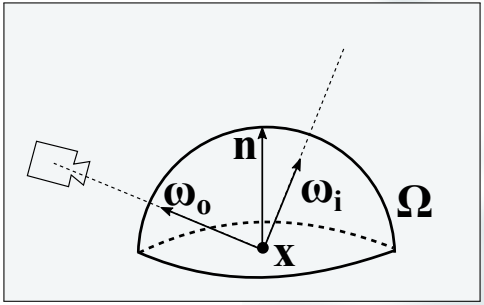
- Macroscopic scale ($>> \lambda$)
- Straight line propagation
- Elastic scattering (static λ)
- Real magnitudes (no complex numbers)

Geometric optics

Effects

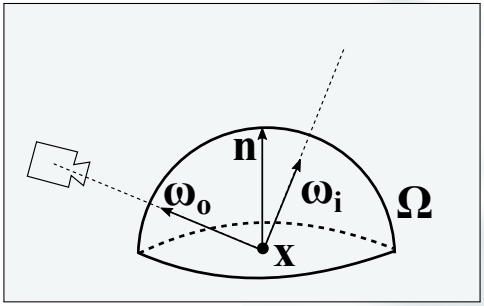
- Emission
- Scattering
- Dispersion
- Absorption

Geometric optics



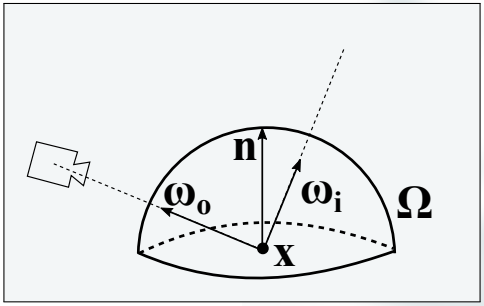
$$L_o(\mathbf{x}, \omega_\mathbf{o}) = L_e(\mathbf{x}, \omega_\mathbf{o}) + \int_\Omega L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\mathbf{n} \cdot \omega_\mathbf{i}| d\omega_\mathbf{i}$$

Geometric optics



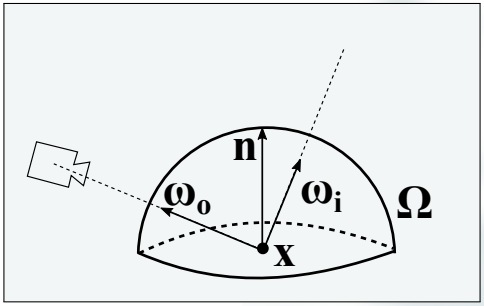
$$L_o(\mathbf{x}, \omega_o) = \underbrace{L_e(\mathbf{x}, \omega_o)}_{\text{EMISSION}} + \int_\Omega \underbrace{L_i(\mathbf{x}, \omega_i)}_{\text{INCOMING LIGHT}} f_r(\mathbf{x}, \omega_i, \omega_o)|\mathbf{n} \cdot \omega_i| d\omega_i$$

Geometric optics



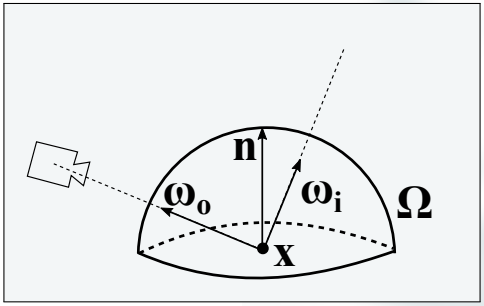
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Geometric optics



$$L_o(\mathbf{x}, \omega_o) = \underbrace{L_e(\mathbf{x}, \omega_o)}_{\text{EMISSION}} + \int_\Omega \underbrace{L_i(\mathbf{x}, \omega_i)}_{\text{INCOMING LIGHT}} \underbrace{f_r(\mathbf{x}, \omega_i, \omega_o)}_{\text{BRDF}} \underbrace{|\mathbf{n} \cdot \omega_i|}_{\text{GEOMETRY}} d\omega_i$$

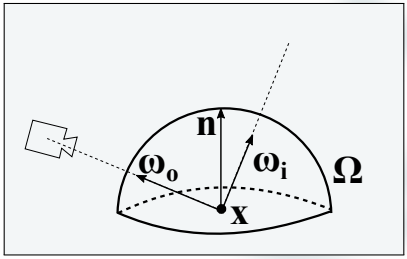
Geometric optics



$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_\Omega \underbrace{L_i(\mathbf{x}, \omega_\mathbf{i})}_{\text{INCOMING LIGHT}} \underbrace{f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o})}_{\text{BRDF}} \underbrace{|\mathbf{n} \cdot \omega_\mathbf{i}|}_{\text{GEOMETRY}} d\omega_\mathbf{i}$$

Radiometry

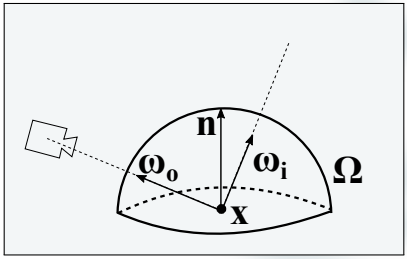
Solid angle



How do we calculate this?

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_\Omega L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\mathbf{n} \cdot \omega_\mathbf{i}| d\omega_\mathbf{i}$$

Solid angle



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What is Ω (integration range for $\omega_\mathbf{i}$)?

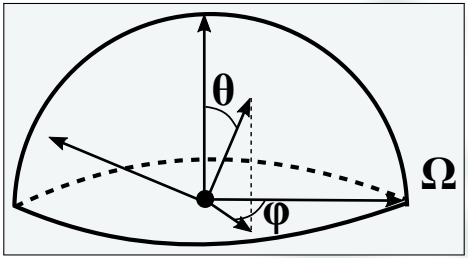
Solid angle

Solid angle: Two-dimensional angle in three-dimensional space that an object subtends at a point. It is a measure of how large the object appears to an observer looking from that point. An object's solid angle in steradians is equal to the area of the segment of a unit sphere, centered at the angle's vertex

Solid angle

1D	2D
Length	Area
Angle	Solid angle

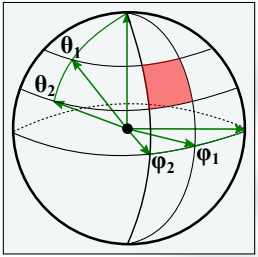
Solid angle



Parametrization

- **Inclination (θ):** Angle with respect to the normal (or any reference vector).
- **Azimuth (ϕ):** Angle within the plane with respect to a tangent vector.

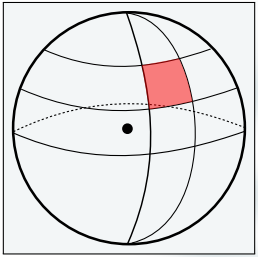
Solid angle



Example: Solid angle patch

$$\omega = |\sin\theta_2 - \sin\theta_1||\phi_2 - \phi_1|$$

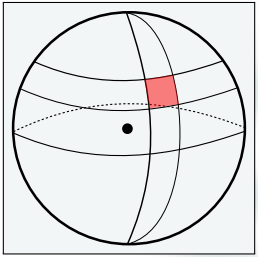
Solid angle



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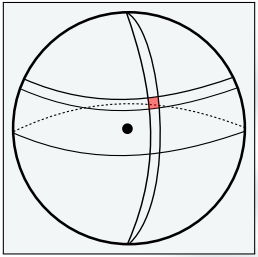
Solid angle



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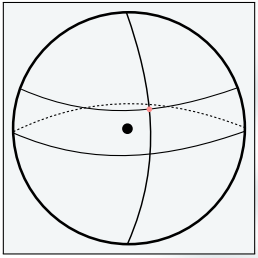
Solid angle



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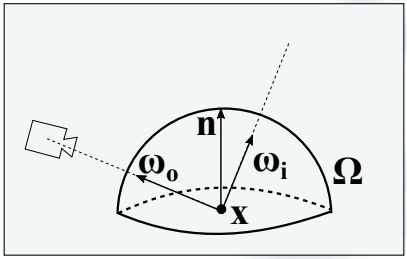
Solid angle



Example: differential solid angle

$$d\omega = |\sin\theta| d\theta d\phi$$

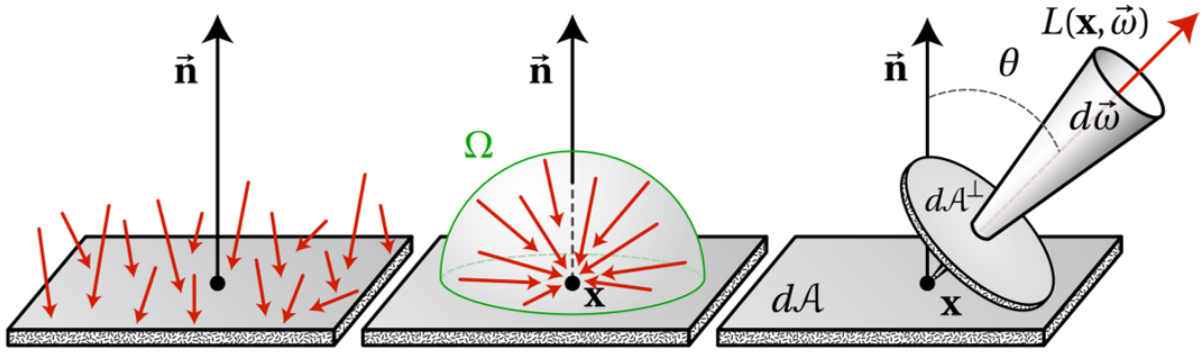
Solid angle



Parametrized:

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\cos\theta_i| |\sin\theta_i| d\theta_i d\phi_i$$

$$\Phi_{3D}(A) = \frac{dQ(A)}{dt} \qquad E_{3D}(\mathbf{x}) = \frac{d\Phi_{3D}(A)}{da(\mathbf{x})} \qquad L_{3D}(\mathbf{x}, \vec{\omega}) = \frac{d^2\Phi_{3D}(A)}{(\vec{\mathbf{n}}\cdot\vec{\omega})^+ d\vec{\omega}\, da(\mathbf{x})}$$



(a) 3D Flux (b) 3D Irradiance (c) 3D Radiance

[Jarosz 2012]

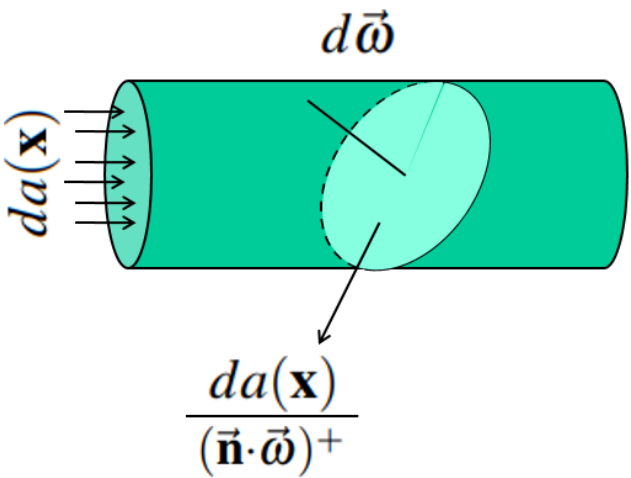
Radiance

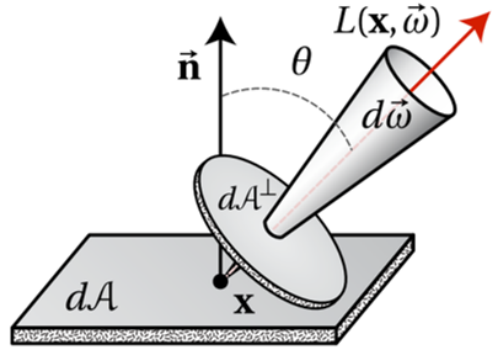
Why is it important?

- It is the magnitude related by the render equation.
- Invariant through a straight line (in the void)
- The sensor's response depends on radiance.

Reminder: It can be expressed as a function of wavelength.

$$L_{3D}(\mathbf{x}, \vec{\omega}) = \frac{d^2\Phi_{3D}(A)}{(\vec{\mathbf{n}}\cdot\vec{\omega})^+ \, d\vec{\omega}\, da(\mathbf{x})}$$

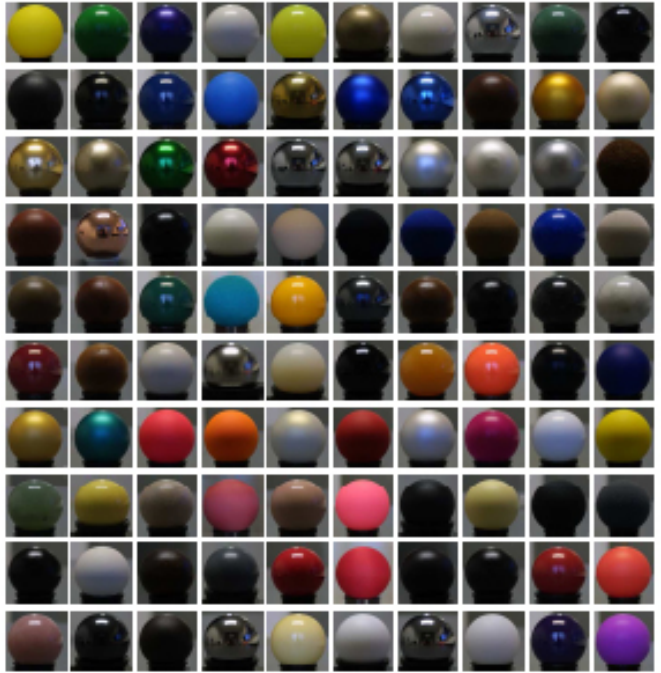




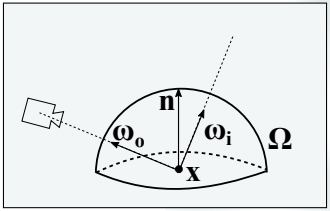
(c) 3D Radiance

[Jarosz 2012]

Reflectance



Reflectance



$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_\Omega L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\mathbf{n} \cdot \omega_\mathbf{i}| d\omega_\mathbf{i}$$

What is $f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o})$?

Reflectance

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_\Omega L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\mathbf{n} \cdot \omega_\mathbf{i}| d\omega_\mathbf{i}$$

What is $f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o})$?

- Bidirectional Scattering Distribution Function (BSDF).
- Scattering = Reflectance on one side (BRDF).
- Scattering = Transmittance on the other side (BTDF).

Properties

Energy conservation

$$\int_{\Omega} f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\mathbf{n} \cdot \omega_\mathbf{i}| d\omega_\mathbf{i} \leq 1$$

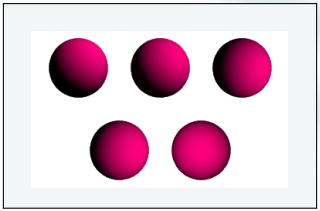
Reciprocity:

$$\int_{\Omega} f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) = \int_{\Omega} f_r(\mathbf{x}, \omega_\mathbf{o}, \omega_\mathbf{i})$$

Lambertian

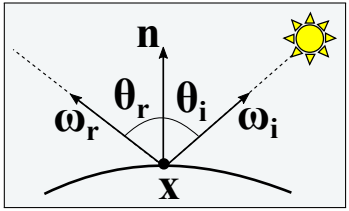
$$f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) = \frac{k_d}{\pi}$$

k_d is the *diffuse coefficient*.
Lambertian BRDF is constant. Therefore lighting depends only on the cosine and it is independent from the point of view.



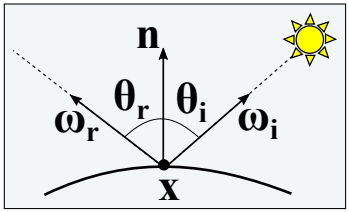
Perfect specular

Law of reflection



Perfect specular

Law of reflection



$$\theta_i = \theta_r$$

$$\omega_\mathbf{r} = \omega_\mathbf{i} - 2\mathbf{n}\left(\omega_\mathbf{i} \cdot \mathbf{n}\right)$$

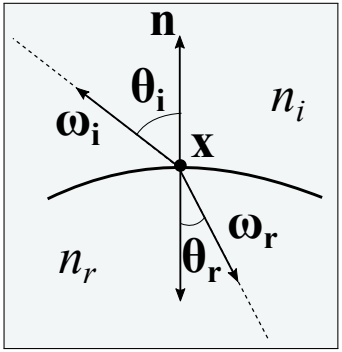
Perfect specular



$$f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) = \frac{\delta_{\omega_\mathbf{r}}(\omega_\mathbf{i})}{(\omega_\mathbf{i} \cdot \mathbf{n})}$$

A delta function δ is 1 for a specific value and 0 for the rest of potential values.

Perfect refraction



$$n_i \sin\theta_i = n_r \sin\theta_r$$

$$f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{r}) = \frac{\delta_{\omega_\mathbf{r}}(\omega_\mathbf{i})}{(\omega_\mathbf{i} \cdot \mathbf{n})}$$


ROYAL TALENS

Direct vs Global Illumination


Direct Illumination

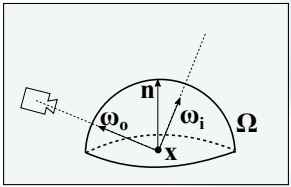

Global Illumination

Direct vs Global Illumination





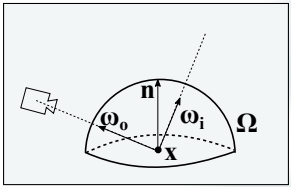
Direct lighting



How do we calculate this?

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_\Omega L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\mathbf{n} \cdot \omega_\mathbf{i}| d\omega_\mathbf{i}$$

Direct lighting

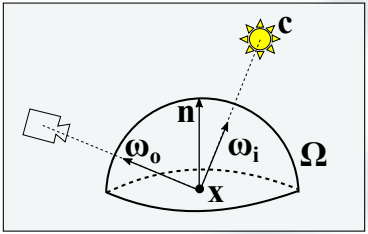


How do we calculate this?

$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \int_\Omega L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\mathbf{n} \cdot \omega_\mathbf{i}| d\omega_\mathbf{i}$$

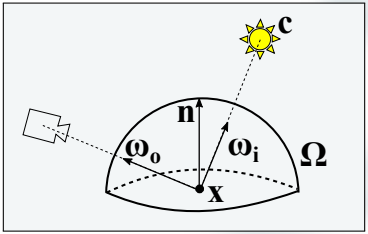
At least we know how to calculate direct light (delta function). Do we?

Direct lighting



$$L_o(\mathbf{x}, \omega_\mathbf{o}) = L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\mathbf{n} \cdot \omega_\mathbf{i}|$$

Direct lighting

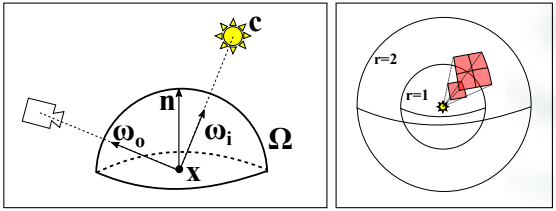


$$L_o(\mathbf{x}, \omega_\mathbf{o}) = L_i(\mathbf{x}, \omega_\mathbf{i}) f_r(\mathbf{x}, \omega_\mathbf{i}, \omega_\mathbf{o}) |\mathbf{n} \cdot \omega_\mathbf{i}|$$

What is $L_i(\mathbf{x}, \omega_\mathbf{i})$?

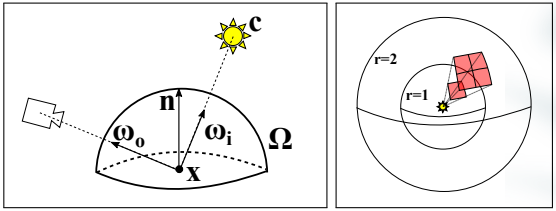
Direct lighting

Point lights



Direct lighting

Point lights

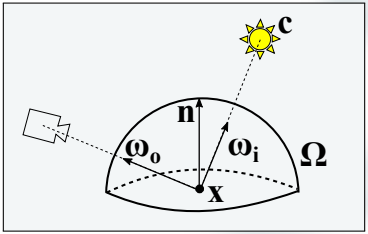


For a point light in position $\mathbf{c}$ with power p:

$$\omega_\mathbf{i} = \frac{\mathbf{c} - \mathbf{x}}{|\mathbf{c} - \mathbf{x}|}$$

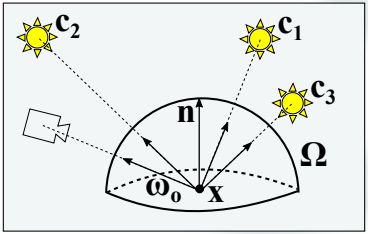
$$L_i(\mathbf{x}, \omega_\mathbf{i}) = \frac{p}{|\mathbf{c} - \mathbf{x}|^2}$$

Direct lighting



$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \frac{p}{|\mathbf{c}-\mathbf{x}|^2} f_r\left(\mathbf{x}, \frac{\mathbf{c}-\mathbf{x}}{|\mathbf{c}-\mathbf{x}|}, \omega_\mathbf{o}\right) \left|\mathbf{n} \cdot \frac{\mathbf{c}-\mathbf{x}}{|\mathbf{c}-\mathbf{x}|}\right|$$

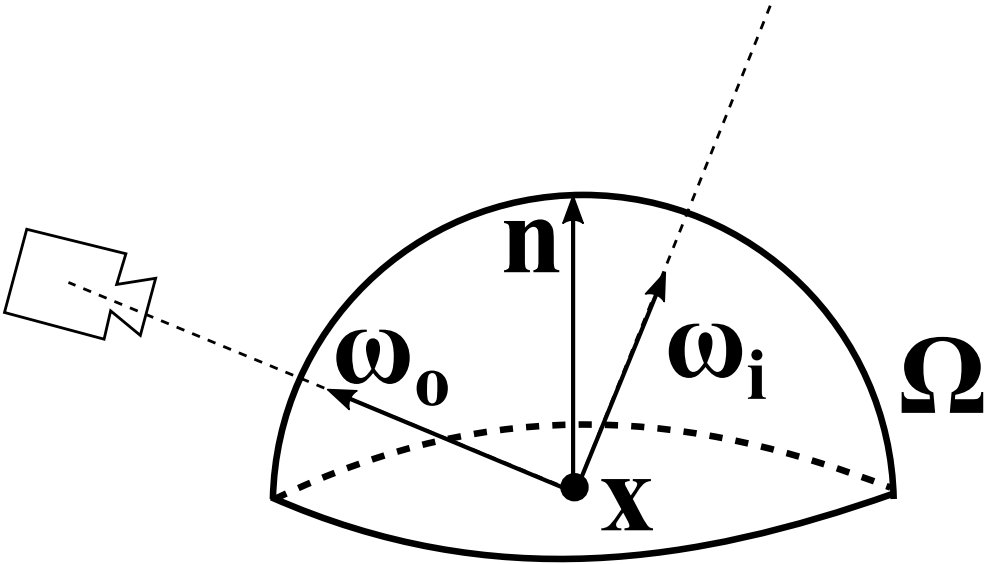
Direct lighting

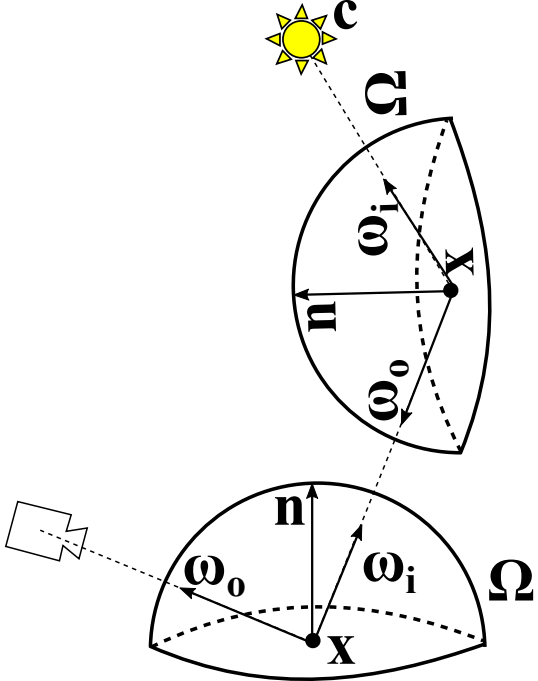


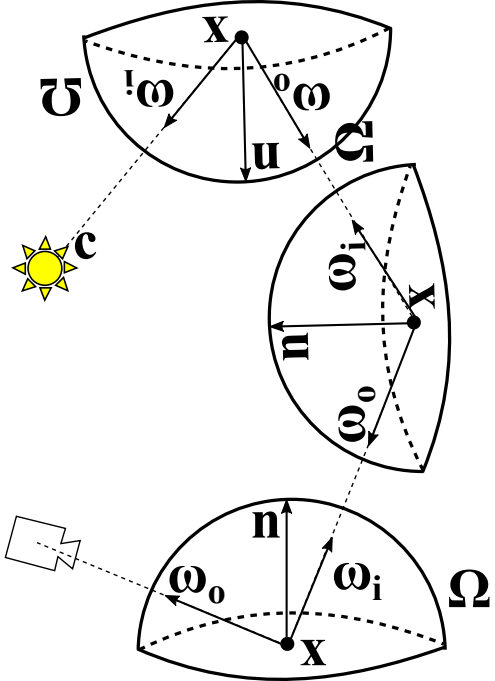
$$L_o(\mathbf{x}, \omega_\mathbf{o}) = \sum_{i=1}^{n} \frac{p_i}{|\mathbf{c_i} - \mathbf{x}|^2} f_r\left(\mathbf{x}, \frac{\mathbf{c_i} - \mathbf{x}}{|\mathbf{c_i} - \mathbf{x}|}, \omega_\mathbf{o}\right) \left|\mathbf{n} \cdot \frac{\mathbf{c_i} - \mathbf{x}}{|\mathbf{c_i} - \mathbf{x}|}\right|$$

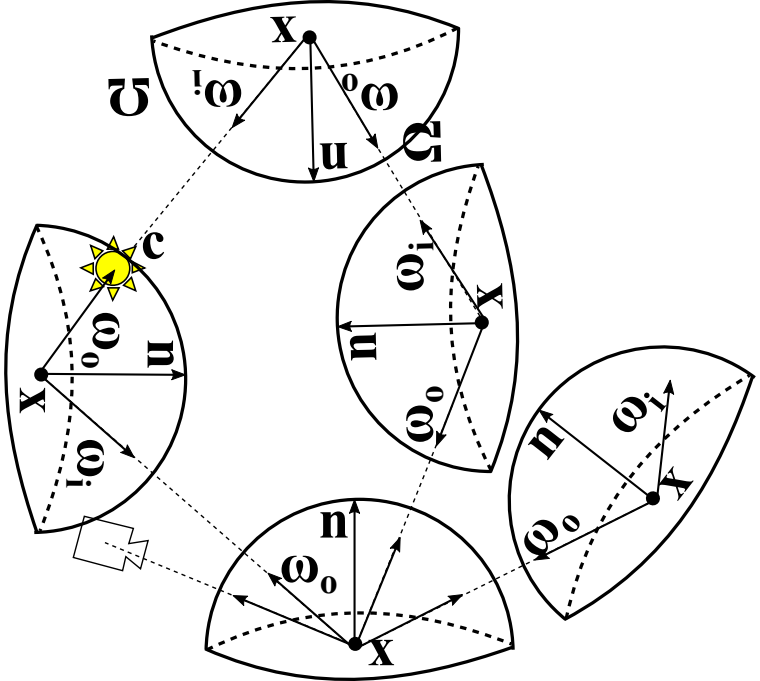

Direct Illumination

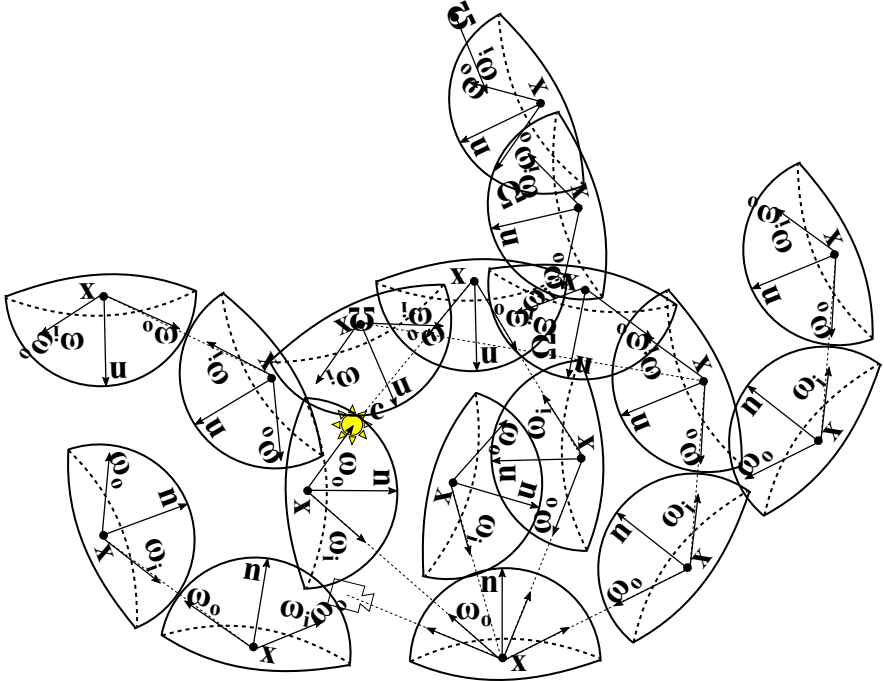

Global Illumination


n
ω_o
ω_i
Ω
x


c
Ω'
ωi
x
n
ωo
n
ωo
ωi
Ω
x


x
u
c
y
n
n
x


x
u
c
x
u
x
n
x
n
x
n
x


x
u
n
e

Global illumination

One bounce

$$L_o(\mathbf{x_1}, \omega_{\mathbf{o1}}) = \int_{\Omega_1} L_i(\mathbf{x_1}, \omega_{\mathbf{i1}}) f_r(\mathbf{x_1}, \omega_{\mathbf{i1}}, \omega_{\mathbf{o1}}) |\mathbf{n_1} \cdot \omega_{\mathbf{i1}}| d\omega_{\mathbf{i1}}$$

Global illumination

One bounce

$$L_o(\mathbf{x_1}, \omega_{\mathbf{o1}}) = \int_{\Omega_1} L_i(\mathbf{x_1}, \omega_{\mathbf{i1}}) f_r(\mathbf{x_1}, \omega_{\mathbf{i1}}, \omega_{\mathbf{o1}}) |\mathbf{n_1} \cdot \omega_{\mathbf{i1}}| d\omega_{\mathbf{i1}}$$

Two bounces

$$L_o(\mathbf{x_1}, \omega_{\mathbf{o1}}) = \int_{\Omega_1} \left(\int_{\Omega_2} L_i(\mathbf{x_2}, \omega_{\mathbf{i2}}) f_r(\mathbf{x_2}, \omega_{\mathbf{i2}}, -\omega_{\mathbf{i1}}) |\mathbf{n_2} \cdot \omega_{\mathbf{i2}}| d\omega_{\mathbf{i1}} \right) f_r(\mathbf{x_1}, -\omega_{\mathbf{i1}}, \omega_{\mathbf{o1}}) |\mathbf{n_1} \cdot \omega_{\mathbf{i1}}| d\omega_{\mathbf{i1}}$$

Global illumination

Three bounces

$$L_o(\mathbf{x_1}, \omega_{\mathbf{o1}}) = \int_{\Omega_1} \left(\int_{\Omega_2} \left(\int_{\Omega_3} L_i(\mathbf{x_3}, \omega_{\mathbf{i3}}) f_r(\mathbf{x_3}, \omega_{\mathbf{i3}}, -\omega_{\mathbf{i2}}) |\mathbf{n_3} \cdot \omega_{\mathbf{i3}}| d\omega_{\mathbf{i3}} \right) \right.$$

$$f_r(\mathbf{x_2}, \omega_{\mathbf{i2}}, -\omega_{\mathbf{i1}}) |\mathbf{n_2} \cdot \omega_{\mathbf{i2}}| d\omega_{\mathbf{i1}} \Big)$$

$$f_r(\mathbf{x_1}, -\omega_{\mathbf{i1}}, \omega_{\mathbf{o1}}) |\mathbf{n_1} \cdot \omega_{\mathbf{i1}}| d\omega_{\mathbf{i1}}$$

Global illumination

N bounces

$$L_o(\mathbf{x_1}, \omega_{\mathbf{o1}}) = \int_{\Omega_1} \cdots \int_{\Omega_N} L_i(\mathbf{x_N}, \omega_{\mathbf{N2}}) \left(\prod_{k=2}^{N} f_r\left(\mathbf{x_k}, \omega_{\mathbf{il}}, -\omega_{\mathbf{i(k-1)}}\right) \right) f_r(\mathbf{x_1}, \omega_{\mathbf{i1}}, \omega_{\mathbf{o1}})$$

$$\left(\prod_{k=1}^{N} |\mathbf{n_k} \cdot \omega_{\mathbf{ik}}| \right) d\omega_{\mathbf{iN}} \cdots d\omega_{\mathbf{i1}}$$

Global illumination

N bounces

$$L_o(\mathbf{x_1}, \omega_{\mathbf{o1}}) = \int_{\Omega_1} \cdots \int_{\Omega_N} L_i(\mathbf{x_N}, \omega_{\mathbf{N2}}) \left(\prod_{k=2}^{N} f_r\left(\mathbf{x_k}, \omega_{\mathbf{il}}, -\omega_{\mathbf{i(k-1)}}\right) \right) f_r(\mathbf{x_1}, \omega_{\mathbf{i1}}, \omega_{\mathbf{o1}})$$

$$\left(\prod_{k=1}^{N} |\mathbf{n_k} \cdot \omega_{\mathbf{ik}}| \right) d\omega_{\mathbf{iN}} \cdots d\omega_{\mathbf{i1}}$$

This will become the **PATH INTEGRAL** (more about it later in the course)


Direct Illumination


Global Illumination

Lighting

Informática Gráfica



Adolfo Muñoz – Julio Marco

Grado en Ingeniería Informática

Graphics and Imaging Lab